SilkSphere

YOLOv8 Silkworm Disease Detection — Model Evaluation Report

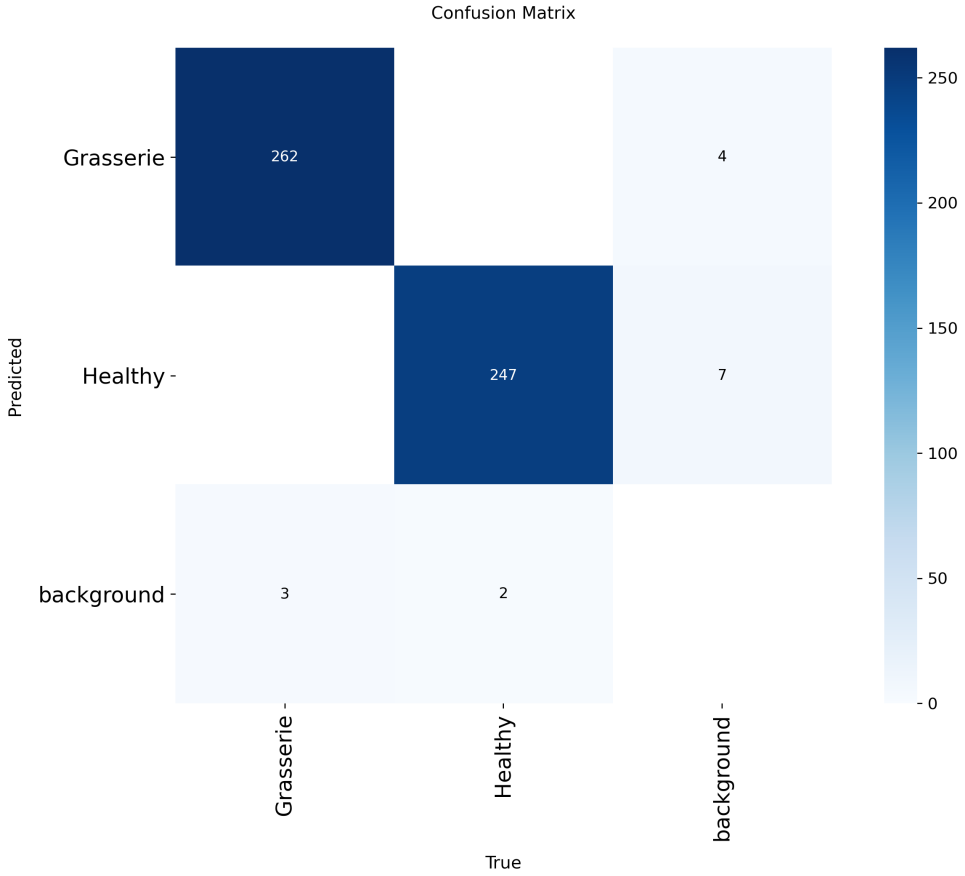
Evaluation charts from YOLOv8 training (88 epochs) on the Roboflow-prepared dataset, for two classes: **Healthy** and **Grasserie**.

Metric	Value
mAP@0.5 (all classes)	0.991
mAP@0.5 — Grasserie / Healthy	0.990 / 0.992
mAP@0.5:0.95	≈ 0.70
Precision / Recall (final)	≈ 0.98 / ≈ 0.99
Best F1 (all classes)	0.99 at confidence 0.604
Precision = 1.00 at confidence	0.855
Correct detections (validation)	262 Grasserie, 247 Healthy

Summary

The model detects and separates healthy and Grasserie-infected silkworms with very high accuracy. No Grasserie/Healthy confusions appear in the confusion matrix, training converged cleanly, and a confidence threshold around 0.5–0.6 gives the best balance of precision and recall for disease counting.

1. Confusion Matrix



WHAT IT SHOWS

Raw counts of predictions vs. ground truth on the validation set.

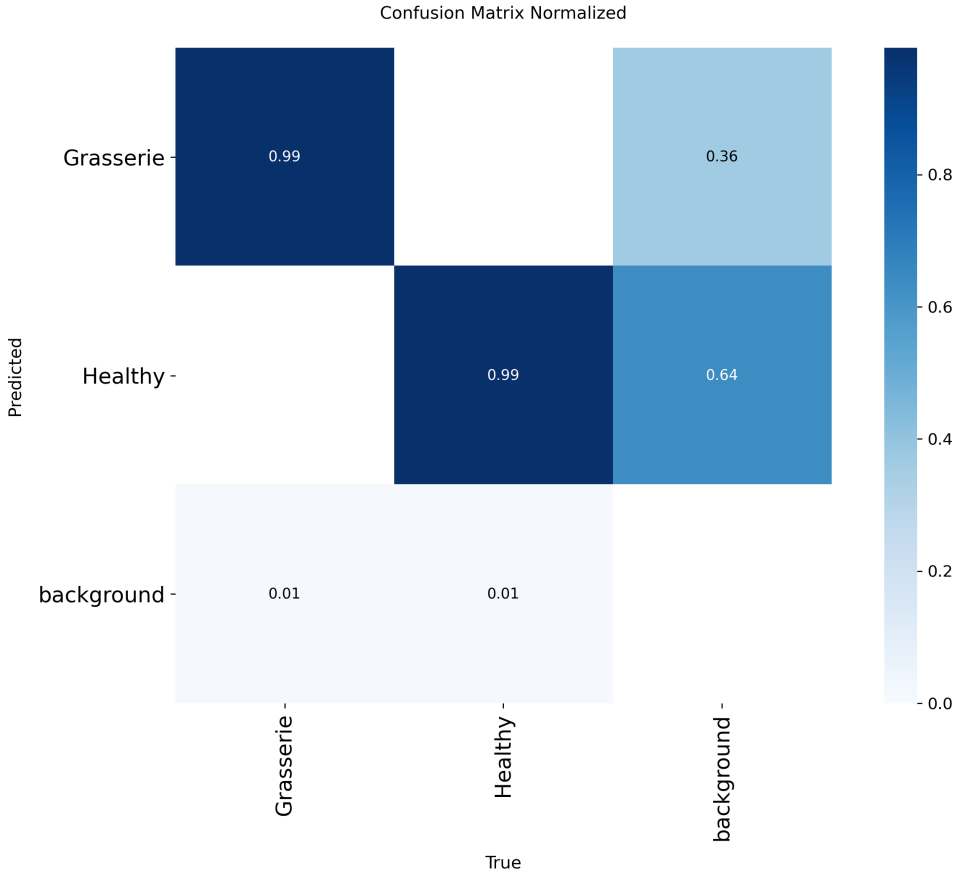
WHAT THE RESULT SAYS

Grasserie: 262 correct, 3 missed (predicted as background), 4 false detections on background. Healthy: 247 correct, 2 missed, 7 false detections on background. There is no Grasserie↔Healthy mix-up at all.

TAKEAWAY

Class-wise precision ≈ 98.5% (Grasserie) and 97.2% (Healthy); recall ≈ 98.9% and 99.2%.

2. Confusion Matrix (Normalized)



WHAT IT SHOWS

The same matrix as per-column proportions of each true class.

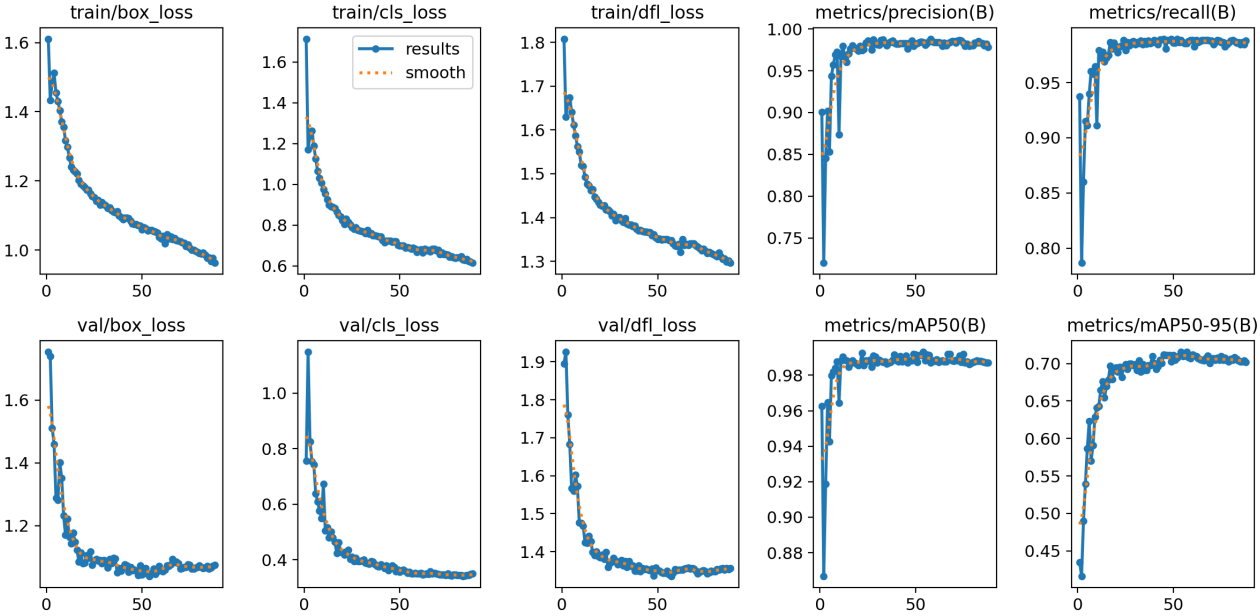
WHAT THE RESULT SAYS

99% of true Grasserie and 99% of true Healthy silkworms are detected correctly; only 1% of each is missed. The background column (0.36 / 0.64) means the few false detections lean toward “Healthy”.

TAKEAWAY

The model almost never confuses a diseased silkworm with a healthy one — essential for reliable disease counting.

3. Training Results (88 epochs)



WHAT IT SHOWS

Losses and metrics over training: box, class and DFL loss (train & validation), plus precision, recall, mAP50 and mAP50-95.

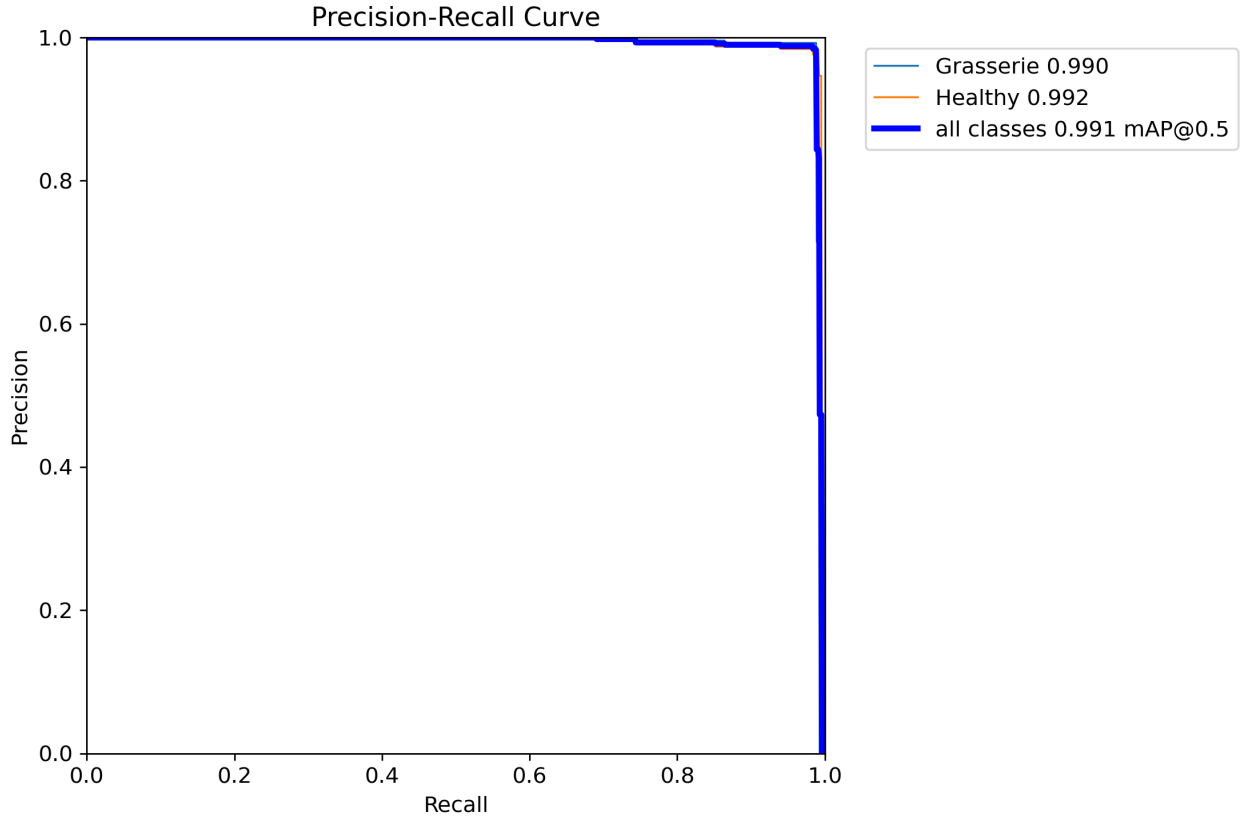
WHAT THE RESULT SAYS

Training and validation losses fall steadily and flatten, with no sign of validation loss rising (no clear overfitting). Precision ≈ 0.98, recall ≈ 0.99, mAP50 ≈ 0.99 and mAP50-95 ≈ 0.70 at convergence.

TAKEAWAY

Training converged smoothly; the model is stable and well fitted to the dataset.

4. Precision–Recall Curve



WHAT IT SHOWS

Trade-off between precision and recall across all thresholds; area under it is mAP@0.5.

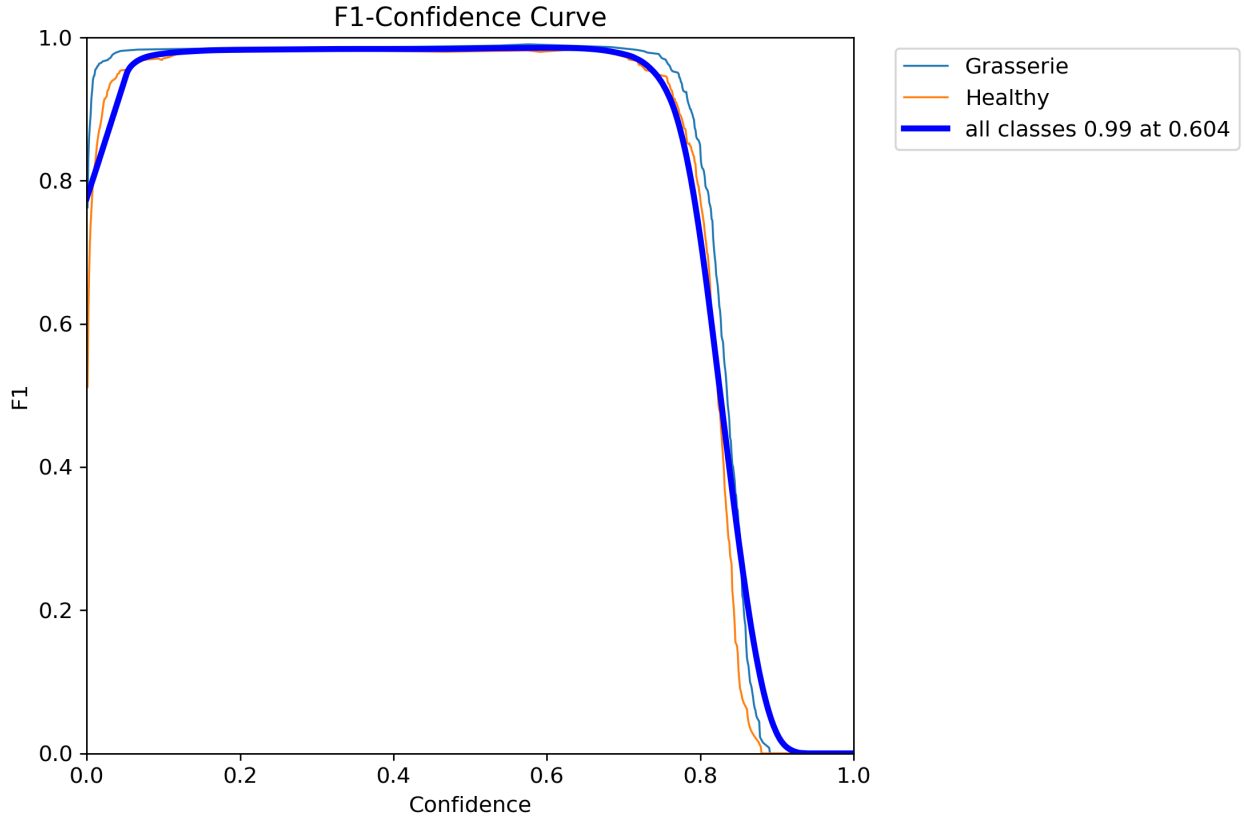
WHAT THE RESULT SAYS

Grasserie AP = 0.990, Healthy AP = 0.992, all classes mAP@0.5 = 0.991. Curves hug the top-right corner.

TAKEAWAY

Near-perfect detection quality for both classes: high precision is kept up to very high recall.

5. F1–Confidence Curve



WHAT IT SHOWS

F1 score (balance of precision and recall) at each confidence threshold.

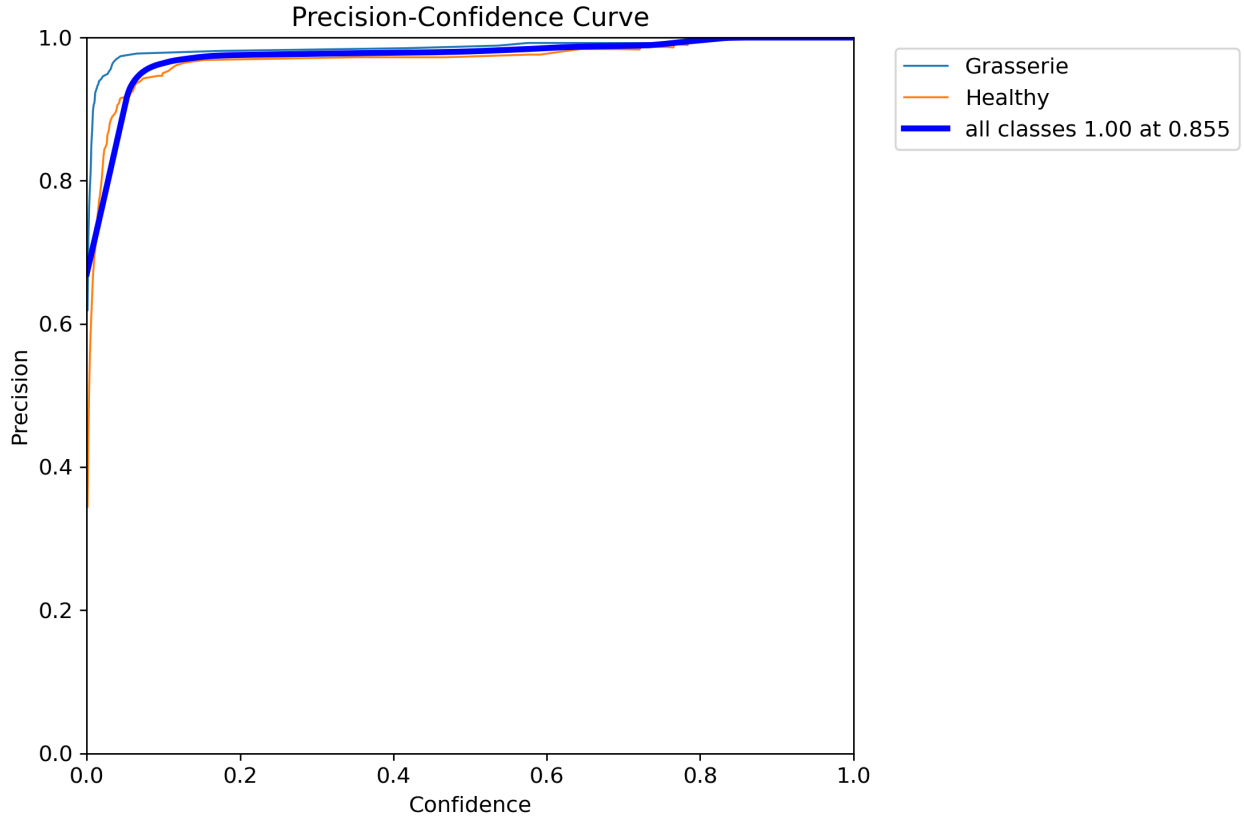
WHAT THE RESULT SAYS

The plot legend reports the best F1 for all classes as 0.99 at a confidence of 0.604. F1 stays near 0.98–0.99 across a wide range (about 0.1 to 0.7), then drops beyond ~0.75.

TAKEAWAY

A confidence threshold of about 0.5–0.6 is a good operating point for SilkSphere's counting.

6. Precision–Confidence Curve



WHAT IT SHOWS

Precision as the confidence threshold rises.

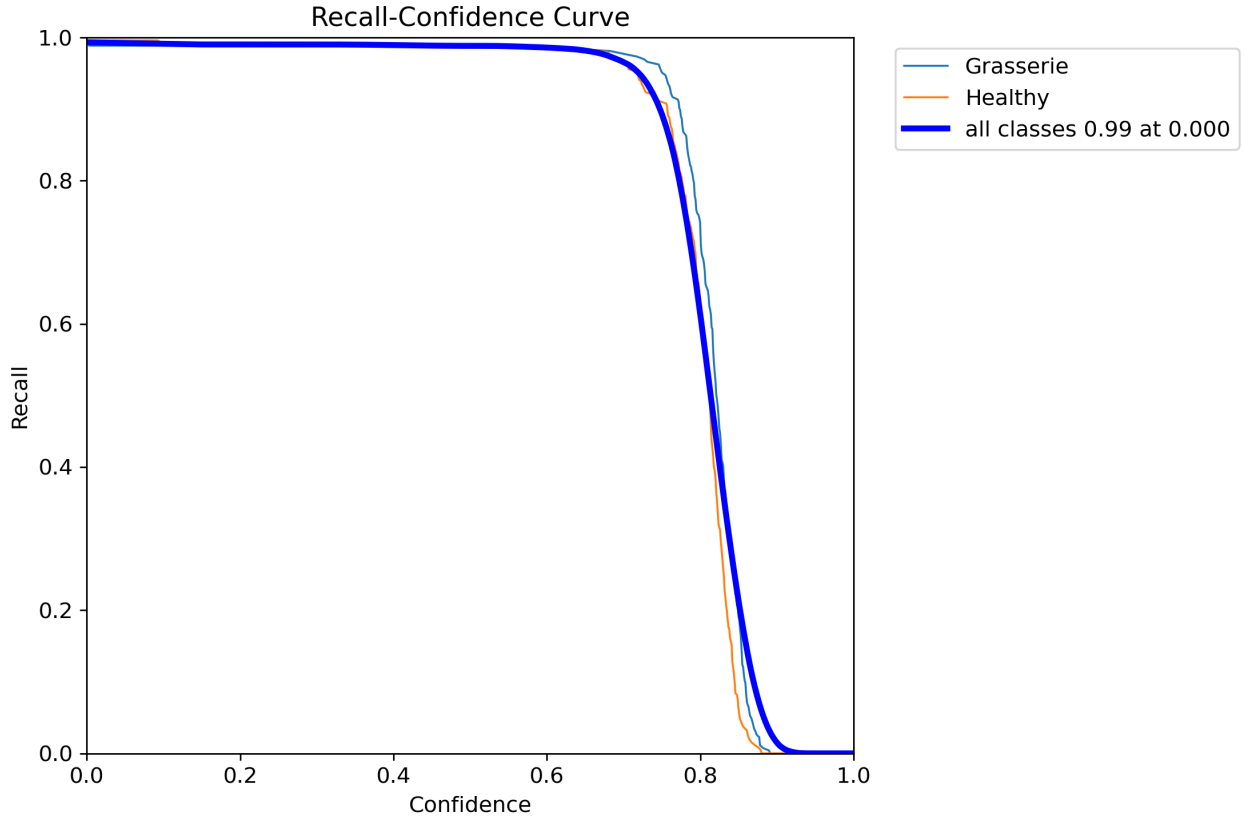
WHAT THE RESULT SAYS

Precision exceeds 0.95 at low thresholds and reaches 1.00 at a confidence of 0.855 for all classes. Grasserie is slightly more precise than Healthy at low confidence.

TAKEAWAY

Raising the threshold removes false positives; at high confidence, predictions are essentially always correct.

7. Recall–Confidence Curve



WHAT IT SHOWS

Recall as the confidence threshold rises.

WHAT THE RESULT SAYS

Recall is about 0.99 at low-to-moderate thresholds and only falls sharply above ~0.75, reaching zero near 0.9.

TAKEAWAY

Avoid very high thresholds (above ~0.75) or real silkworms start being missed. This balances the precision curve above.